System Design

Project Title:

ProjectPilot AI

Problem Statement:

Students and beginner developers often struggle to convert project ideas into executable plans. They face challenges in project decomposition, architecture design, roadmap planning, risk assessment, and task allocation.

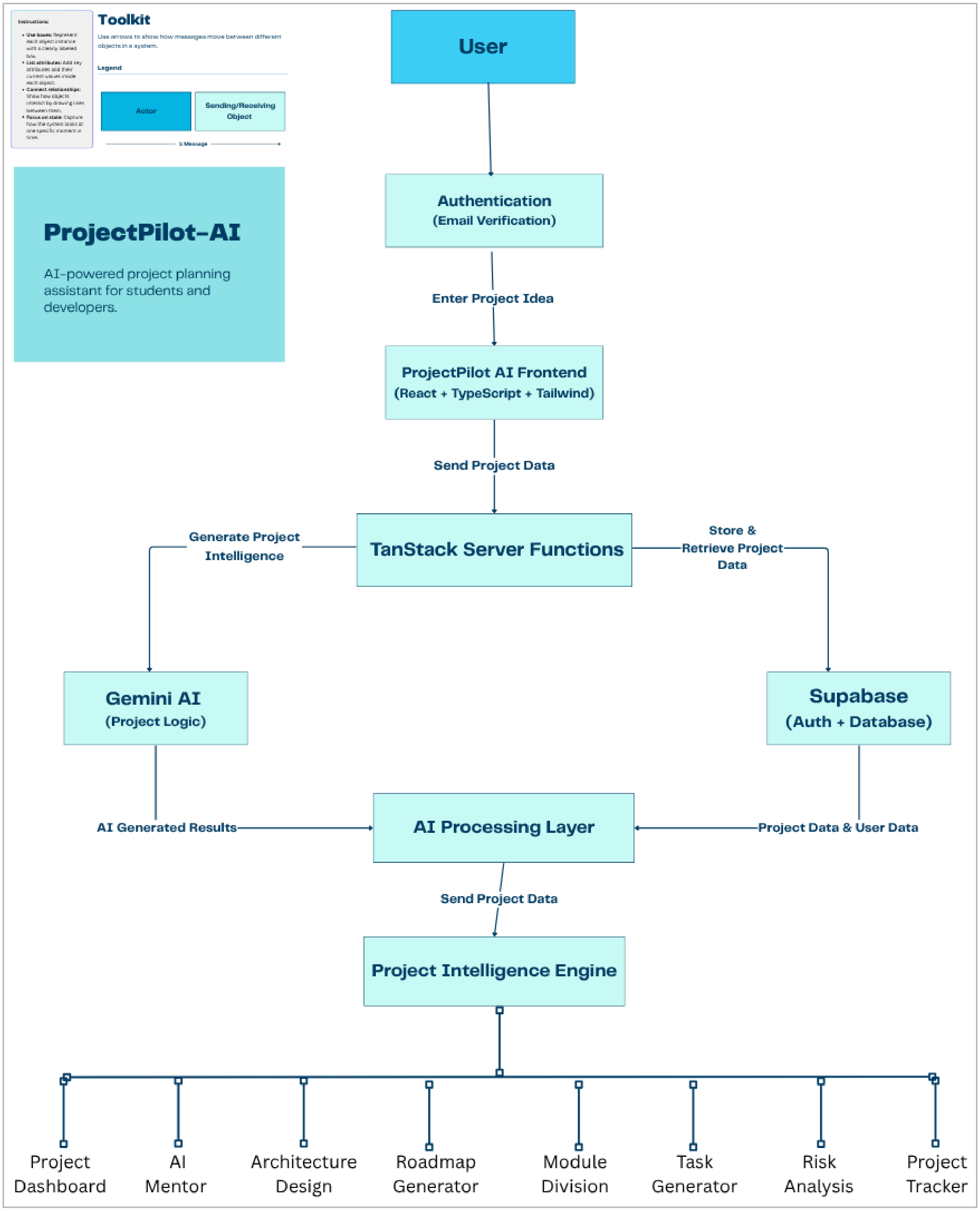
Solution:

ProjectPilot AI is an AI-powered project planning assistant that transforms a simple project idea into a structured execution strategy using Gemini AI.

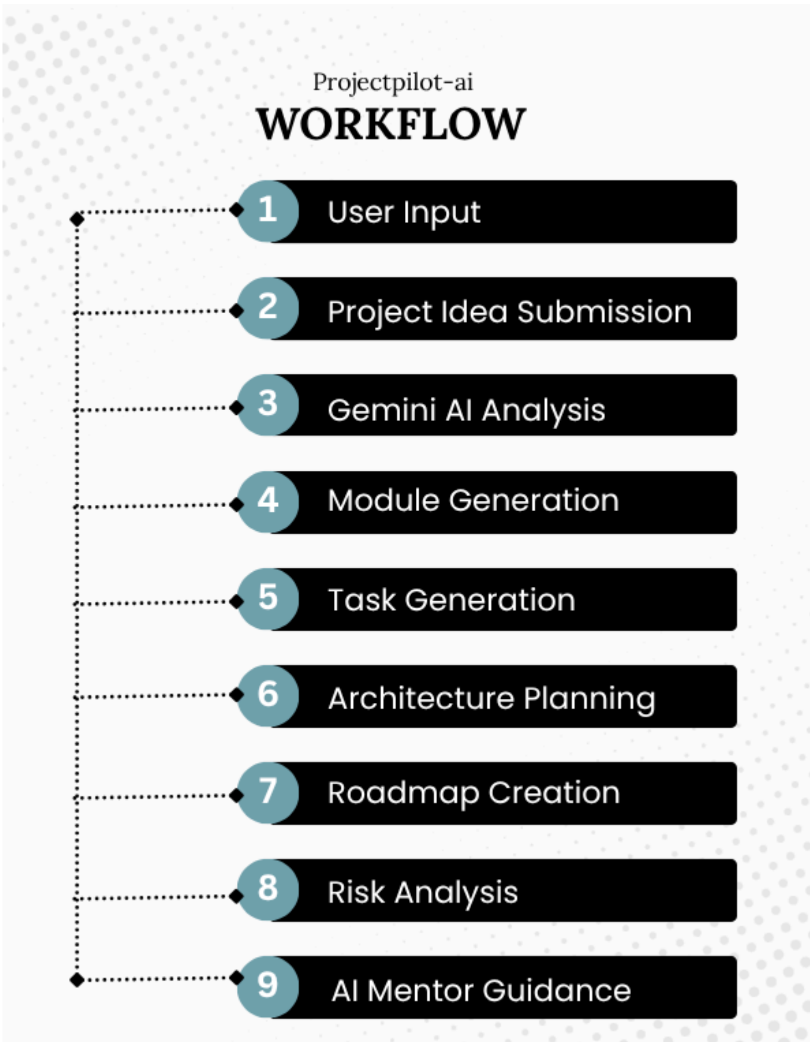
Key Features

- AI Module Generation
- AI Task Generation
- Architecture Design
- Roadmap Planning
- Risk Analysis
- AI Mentor
- Project Simulation
- Progress Tracking

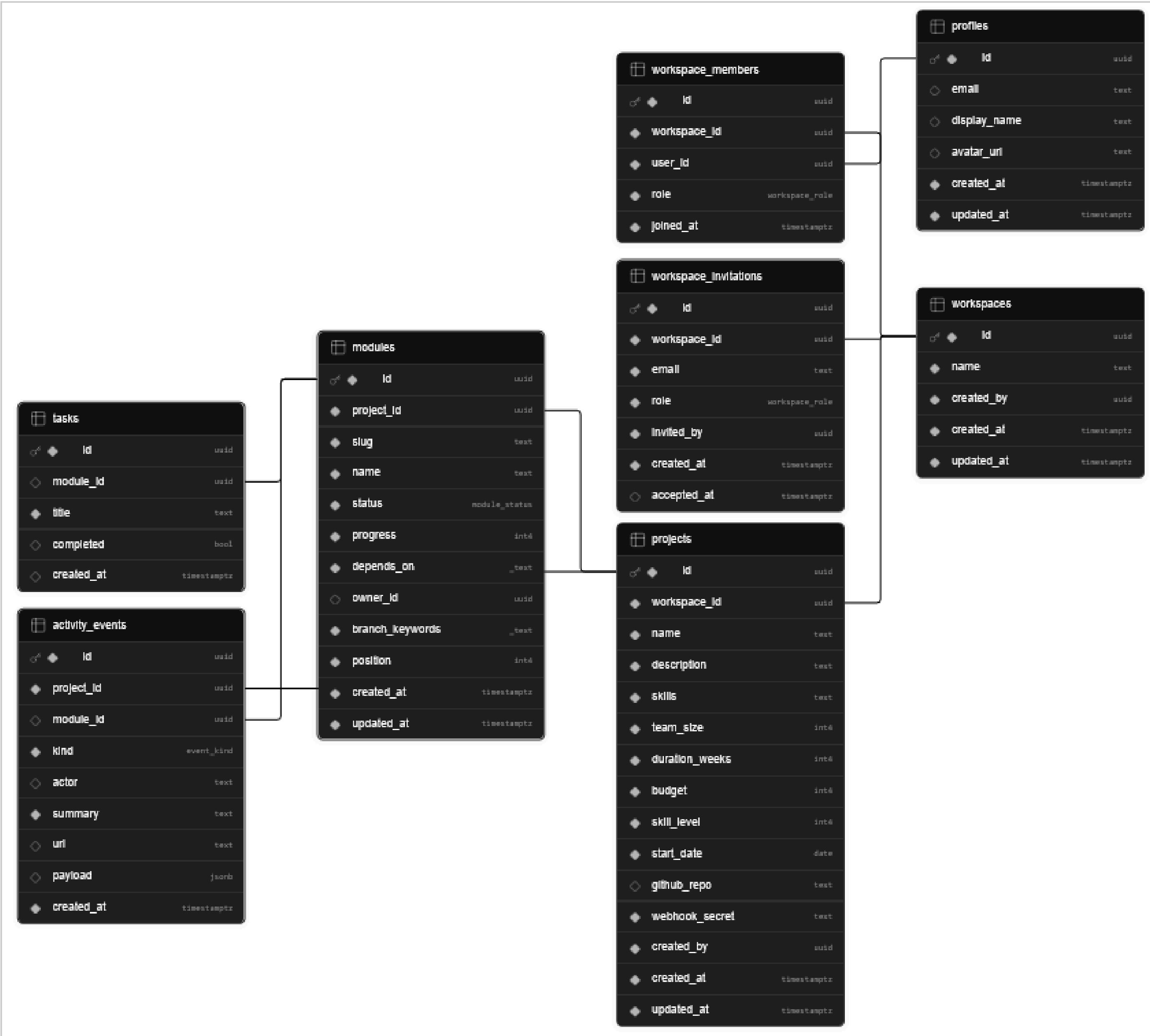
System Architecture :



Workflow:


Projectpilot-ai
WORKFLOW
1 User Input
2 Project Idea Submission
3 Gemini AI Analysis
4 Module Generation
5 Task Generation
6 Architecture Planning
7 Roadmap Creation
8 Risk Analysis
9 AI Mentor Guidance

Database Schema:



Technologies Used

Frontend

- React
- TypeScript
- Tailwind CSS
- TanStack Router

Backend

- TanStack Start Server Functions

Database

- Supabase

Authentication

- Supabase Auth

AI Layer

- Gemini 2.5 Flash Lite

Version Control

- Git & GitHub

Prompt Design

Example Prompt:

Project:

AI-powered learning assistant

Generate:

1. Development modules

2. Implementation tasks

3. Architecture suggestions

4. Project roadmap

5. Risks and mitigations

Provide practical and implementation-focused recommendations.

Trade-offs Considered

- Gemini was chosen due to low cost and fast response time.
- Supabase chosen to reduce backend complexity.
- Server Functions used to securely manage API calls.
- Modular architecture allows future feature expansion.

Future Improvements

- Team collaboration
- Gantt chart generation
- Architecture visualization
- Project export functionality
- Real-time collaboration
- Report generation
- Connecting with Github